MSRIP-Net: Addressing Interpretability and Accuracy Challenges in Aircraft Fine-Grained Recognition of Remote Sensing Images

Zhengxi Guo, Biao Hou, *Senior Member, IEEE*, Xianpeng Guo, Zitong Wu, Chen Yang, Bo Ren, *Member, IEEE*, and Licheng Jiao, *Fellow, IEEE*

***Abstract*— The task of fine-grained aircraft recognition is crucial in the field of remote sensing. Despite some progress achieved by traditional deep learning methods in addressing this challenge, they are often perceived as a "black box," lacking transparent explanations for model decisions. Current interpretable methods based on attention mechanisms, although providing some interpretability, do not align with human thought logic. Therefore, we propose a multiscale rotation-invariant prototype network (MSRIP-Net). Our approach simulates the intuitive reasoning process of humans in identifying objects by segmenting them into multiple components. Importantly, MSRIP-Net has the capability to automatically recognize rigid components on aircraft targets without relying on additional part annotations, using only image-level class labels. In addition, our approach effectively addresses challenges presented by noise, deformations, and multiscale variations in remote sensing targets and has been comprehensively evaluated on datasets FAIR1M1.0 and Rareplane. Our results demonstrate that MSRIP-Net achieves higher accuracy compared with existing fine-grained recognition methods. Furthermore, we provide insights into the model's decision-making process to illustrate the interpretability of our approach.**

***Index Terms*— Fine-grained aircraft classification, interpretability, prototype network, remote sensing imagery, target recognition.**

I. Introduction

With the development of remote sensing technology, the aircraft fine-grained recognition is a crucial task in the interpretation of remote sensing images, which plays a vital role in military and civilian fields, such as mission planning, security, surveillance, and military decision making [1], [2], [3], [4], [5]. Therefore, in practical application scenarios, achieving high-precision classification results alone is no longer sufficient, what is imperative is the utilization of highly interpretable methods.

In recent years, the algorithms based on deep learning have been extensively applied to the task of aircraft fine-grained recognition in remote sensing [6], [7], [8], [9]. Convolutional neural network (CNN), as representatives of these approaches, possess the capability to learn hierarchical features from vast datasets [10]. Utilizing a unified framework for both feature extraction and classification, CNN excels in extracting discriminative deep features from targets [10], [11], [12].

However, the black-box property of CNN makes it exceedingly challenging for us to comprehend the internal workings and decision-making processes of the network [13], [14]. As shown in Fig. 1, CNN models perform inference by learning optimal network weights and providing results to users. However, users are limited to observing the outcomes without insights into the underlying decision-making processes. Interpretable models, in contrast, provide users with a clear understanding of the decision-making process, offering transparent insights into the decision rationale. This transparency instills a higher level of trust in the model, leading to increased confidence in the results. This is particularly crucial in certain application domains, such as military and monitoring, where the transparency and reliability of decision-making processes are paramount.

Hence, interpretability in deep learning has become a highly scrutinized core issue, particularly in critical domains, such as military and surveillance, which have the urgent demand for transparency and reliability in the decision-making processes of deep models.

A. Related Works and Motivation

In recent years, several studies have recognized the significance of the interpretability of fine-grained recognition tasks for remote sensing targets. These studies mainly perform interpretable inference by focusing on some features of the target, which include three branches.

The first branch is class activation maps (CAMs). The methods based on CAM have been widely employed in postinterpretable tasks of deep learning models, enabling human observers to discern which regions of an image the network

Received 16 August 2024; accepted 9 September 2024. Date of publication 11 September 2024; date of current version 26 September 2024. This work was supported in part by the National Natural Science Foundation of China under Grant 62171347, Grant 62101405, Grant 62371373, and Grant 62271377; in part by the 111 Project; in part by the Postdoctoral Fellowship Program of China Postdoctoral Science Foundation (CPSF) under Grant GZC20232036 and Grant GZC20232032; and in part by Shaanxi Province Postdoctoral Research Project under Grant 2023BSHEDZZ168 and Grant 2023BSHYDZZ96. *(Corresponding author: Biao Hou.)*

The authors are with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, and the Joint International Research Laboratory of Intelligent Perception and Computation, Xidian University, Xi'an 710071, China (e-mail: XDU_Gzx@stu.xidian.edu.cn; avcodec@hotmail.com; guoxianpeng@xidian.edu.cn; wuzitong@stu.xidian.edu.cn; yangchen@xidian.edu.cn; asdwer2046@126.com; lchjiao@mail.xidian.edu.cn).

Digital Object Identifier 10.1109/TGRS.2024.3458408

1558-0644 © 2024 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

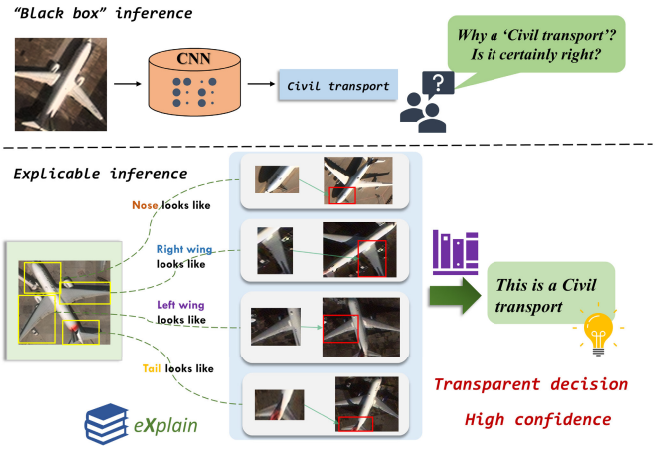


Fig. 1. (Top) Traditional black-box CNN inference and (bottom) interpretable model inference.

focuses on [15], [16], [17], and [18]. Specifically, CAM-based methods utilize features from the last convolutional layer and replace subsequent fully connected layers and softmax layers with a global average pooling (GAP) layer. This GAP layer computes the mean value of all pixels in the feature map to represent the entire feature map and visualizes the most salient results according to the model through a heatmap. However, CAM-based methods inherently do not provide interpretability to the model's decisions, as CAM can only indicate which parts the network attends to after making its decision, rather than elucidating the decision-making process of the model. Furthermore, CAM can only activate region-level areas corresponding to a specific class of objects. Specifically, in the task of fine-grained recognition of aircraft, CAM can only activate regions corresponding to a particular class of aircraft objects, rather than activating different parts of the aircraft.

For the second branch, Shaw et al. [19] introduce self-attention mechanisms to guide the model in focusing on the crucial parts of the target, revealing the key features upon which the model relies during predictions, thereby enhancing the interpretability of the model. For example, Xiong et al. [20] employed attention mechanisms and structural causal models to generate multiple causal attention maps, which increases attention to objects' discriminative parts and explores intrinsic relationships between multiple attention parts and predicted outcomes. Yi et al. [21] employed multilevel attention modules to extract intrinsic features of the target and constructed a feature bilinear aggregation pool for feature fusion. This method aggregated intrinsic features and attention features, obtaining high-level features that are both intraclass coherent and interclass separable. Han et al. [22] designed the dual mask attention module (DMAM) to refine the fused features, enhancing performance in dense and cluttered scenes by accentuating distinctions between targets and suppressing noise. Fu et al. [23] designed a context-aware patch selection module to identify distinctive regions and utilized a self-attention mechanism to aggregate the selected regions. This facilitates adaptive fusion, feature refinement, and optimization of decision making. Pan et al. [24] adopted a causal attention mechanism and employed a "decouple + aggregate" asynchronous feature learning paradigm to enhance the model's capability in fine-grained recognition.

But in fact, the self-attention mechanism just updates the network weight to increase the sensitivity to local areas, which is "pseudointerpretable" [25]. Considering a logical scenario, such as inferring the type of an aircraft, the "attention model" might say, "I determined that it is a 'large passenger plane' based on this region." In contrast, the human reasoning often involves considerations like "I think the nose looks like Boeing series, the two wings look like Boeing-737, and the tail also looks like Boeing-737, so I conclude that it is a Boeing-747 aircraft," which is referred to Fig. 1 (bottom). The self-attention mechanism may overly focus on certain local features of the aircraft, which might not be the key factors human experts or ordinary observers pay attention to when identifying aircraft models. This lack of intuitive interpretability diverges from human perceptual and cognitive logic.

At present, some weakly supervised object detection (WSOD) methods [26], [27], [28], [29] can train an object detector with only image-level class annotations to localize objects, aiming to reduce the cost of manually annotating bounding boxes. However, WSOD methods are primarily tailored for detecting objects within images. Because they operate at the level of entire images rather than individual object parts, they lack the granularity necessary to learn prototypes of different parts of the target and perform fine-grained recognition. In addition, WSOD methods typically focus on identifying the presence or absence of objects in images, rather than extracting detailed structural information or identifying specific object components. Therefore, they are unable to capture the rigid structure of complex objects like aircraft or perform fine-grained recognition tasks that require a detailed understanding of object composition and organization.

Therefore, the third branch is proposed to overcome the above problems. Researchers introduce the additional subnetworks to capture the discriminative semantic parts of fine-grained objects, that is, designing a localization subnetwork (detection or segmentation [30], [31]) to locate key parts, and then obtaining the corresponding part-level feature vectors [32], [33], [34], [35], [36]. The motivation for these models is to first find the corresponding parts and then compare how they look. A typical representative is APPEAR [37], which explicitly models the rigid structure of an aircraft as a pixel-level partwise prior, dividing it into five distinct components: 1) nose; 2) left wing; 3) right wing; 4) fuselage; and 5) tail. Also, use HRNet-Seg [38] to segment different parts and identify subtle interclass differences. This fine-grained part information provides a reliable localization prior for components and delineates the rigid structure of the aircraft, aligning with human perceptual and cognitive logic, achieving interpretability goals.

Although the third branch has achieved interpretability to some extent, both encountered the following issues in practice. Also, they are shown sequentially in Fig. 2.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

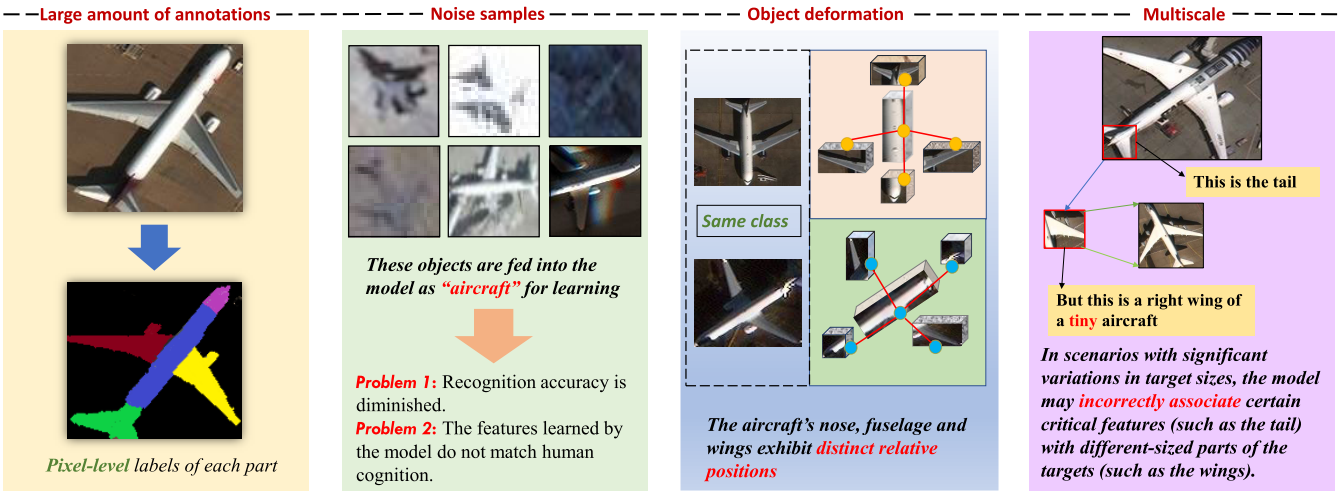


Fig. 2. Challenges faced by existing methods in practical applications.

1) Methods in the third branch often require partial annotation for training, which requires a lot of manual annotation. As mentioned in APPEAR [37], it took three months to label these pixel-level labels of aircraft's parts.
2) The impact of noise samples generated by sensor interference such as cloud and atmospheric disturbances in the training data is overlooked. These noisy samples are treated as "aircraft samples" for model learning, but they do not represent real-world scenarios, contradicting interpretability.
3) Due to the potential variations in the orientation, posture, and position of aircraft targets, the relative spatial positions of their rigid structures can change. Traditional methods often struggle to accurately capture the positional changes of different parts of the target, leading to misjudgments.
4) Due to the multiscale characteristics of remote sensing aircraft targets, there may be an issue of mismatching rigid parts at different scales, incorrectly associating different parts together.

B. Contributions

Therefore, to address the aforementioned issues, this article proposes a multiscale rotation-invariant prototype network (MSRIP-Net), which is the first work to introduce the prototype network in fine-grained aircraft recognition of remote sensing images. Our method emulates the intuitive reasoning process of humans in recognizing objects by segmenting them into multiple components. The model automatically identifies meaningful parts of the aircraft target without relying on additional part annotations with only image-level labels. Furthermore, in view of the characteristics of remote sensing aircraft targets, our method adeptly addresses issues related to noise, deformation, and multiscale variations. Notably, our approach is state-of-the-art (SOTA) accuracy in fine-grained aircraft recognition while achieving a highly interpretable decision-making process. The main contributions are as follows.

1) We propose an MSRIP-Net, which eliminates the need for manually annotating extensive pixel-level labels, relying solely on image-level labels. The method autonomously learns prototypes of each part of every target class (including the nose, left wing, right wing, and tail) given only image-level class labels and retains these learned prototypes within the model. During inference, our method can be conceptualized as a "scoring table" for each category. The network dissects the aircraft target by identifying prototype parts and combines these prototype categories for final classification, providing a transparent decision-making process.
2) In MSRIP-Net, we design a global prior module (GPM), which accurately filters out the noisy samples in the dataset by calculating the difference in the distribution of the aircraft target feature space. It solves the impact of the noisy samples on the accuracy and interpretability of the model and makes the model more focused on the aircraft samples in the real scene.
3) In order to consider the deformation and multiscale characteristics of aircraft targets, we designed a rotation-invariant prototype module (RIPM) and a scale decision module (SDM), which improves the ability of each part of the prototype to match complex shapes and multiscale targets during inference, so that the model can better capture the local structure and deformation of targets.
4) The proposed MSRIP-Net achieves a new SOTA recognition performance on the FAIR1M1.0 and Rareplane fine-grained datasets. Also, we will show the decision-making process on each dataset to prove the interpretability of our method.

The remainder of this article is organized as follows. We describe the background in Section II and introduce our proposed MSRIP-Net in Section III. Evaluations on two public datasets are presented in Section IV, followed by conclusion in Section V.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

II. Background

In this section, we will introduce relevant knowledge of the prototype which is necessary for the rest of this article.

A. Prototype

A prototype is a data instance that represents a set of data. This concept aims to generalize knowledge about an entire dataset or category by using a small number of representative samples. The prototype can be regarded as the characteristic center of a category, and its characteristic values reflect the general properties of the entire category [25], [39], [40], [41], [42]. Selection methods for prototypes include k-medoids, a clustering algorithm, and any algorithm that returns actual data points as cluster centers.

B. Maximum Mean Discrepancy (MMD)

MMD is a statistical metric used to measure the dissimilarity between two probability distributions [43]. It quantifies the discrepancy between the observed data distribution and an idealized distribution, typically represented by a set of prototypes. Given two distribution X and Y with corresponding sample sets $\{x_1, x_2, \ldots, x_m\}$ and $\{y_1, y_2, \ldots, y_n\}$, the MMD is defined as follows:

$$\mathrm{MMD}^2(X,Y) = \frac{1}{m^2}\sum_{i,j=1}^{m} k\left(x_i, x_j\right) - \frac{2}{mn}\sum_{i,j=1}^{m,n} k\left(x_i, y_j\right) + \frac{1}{n^2}\sum_{i,j=1}^{n} k\left(y_i, y_j\right) \tag{1}$$

where $k(\cdot,\cdot)$ is the kernel function.

The performance of MMD depends heavily on the choice of kernel function. Commonly used kernel functions include linear kernel function, polynomial kernel function, and Gaussian kernel function. Later, we will introduce the kernel function selected in this article to calculate the aircraft target. MMD is an unsupervised method that does not require class labels, and unlike kernel methods that map to high-dimensional spaces, MMD is calculated directly in the original feature space, avoiding the curse of dimensionality problems. Please refer to [43] for the details.

III. Method

As shown in Fig. 3, our method consists of four key modules, the first is the GPM, which is responsible for preprocessing data. The main task of GPM is to offline filter and remove noise samples from the dataset for improving data quality. It is worth noting that GPM does not participate in the calculation of the model during the inference stage. The second part is the feature extractor. We chose ResNet-18 [44] as the backbone network for feature extraction. The third part is the SDM, whose main function is to use the pooling layer for multiscale feature extraction. This helps provide the prototype layer with information at different scales for better matching. A key part of the network is the RIPM, which is designed to achieve invariance to rotation and make the model more robust. This module is responsible for matching feature maps with prototypes during inference to classify the target. Finally, we introduce the loss function and training strategy used to learn the prototypes of each component.

A. Global Prior Module

To enhance the interpretability of the dataset, we propose a GPM based on MMD. This module effectively identifies and eliminates outlier samples that deviate from the overall data distribution, thereby improving the quality and performance of model training. It is important to note that GPM operates as an offline module, functioning exclusively during the data preprocessing phase and not participating in the subsequent training process. The high-quality dataset filtered by GPM allows the model to capture the essential characteristics of the data more accurately.

Initially, we standardize the size of all aircraft targets to 32×32 and flatten into 1×1024 pixel value vectors, denoted by v_i. Subsequently, each vector undergoes min–max normalization independently as follows:

$$v_i = \frac{2 \cdot (v_i - v_i)}{\min - \max} - 1. \tag{2}$$

For calculating the MMD between two samples, we opt for the Gaussian kernel function (radial basis function), as depicted in the following equation:

$$k(x,y) = \exp\left(-\frac{\|x-y\|^2}{2\sigma^2}\right). \tag{3}$$

This kernel function possesses rotation-invariant properties, making it suitable for our task of noise sample filtration. Since this part of the task focuses on filtering out noise samples without considering scale and rotation, the selected samples are expected to be more accurate and less prone to errors caused by differences in angles and scales.

1) Prototype Samples Selection: First, it is essential to select a set of prototype samples that can best represent the entire aircraft dataset. We randomly choose m samples from the entire dataset as the initial prototype set $\{p_1, p_2, \ldots, p_m\}$. For each p, calculate the MMD^2 value concerning the aircraft dataset $\{v_1, v_2, \ldots, v_n\}$, which measures the difference in distribution between the prototype sample and the entire dataset

$$\mathrm{MMD}^2 = \frac{1}{m^2}\sum_{i,j=1}^{m} k_{\mathrm{gauss}}\left(p_i, p_j\right) - \frac{2}{mn}\sum_{i,j=1}^{m,n} k_{\mathrm{gauss}}\left(p_i, v_j\right) + \frac{1}{n^2}\sum_{i,j=1}^{n} k_{\mathrm{gauss}}\left(v_i, v_j\right) \tag{4}$$

where k_{gauss} is the Gaussian kernel function that measures the similarity of two samples, m is the number of prototypes, and n is the sample number of the original dataset.

Our goal is to minimize MMD^2. A lower MMD^2 value indicates a better fit between the distribution of prototypes and the original data distribution. We use greedy search to select prototypes with an empty list of prototypes, while the number of prototypes is below the chosen number m. For each sample in the dataset, assess how much MMD^2 is reduced when the point is added to the list of prototypes. Also, add the sample that minimizes the MMD^2 to the list, which contains the prototypes of the dataset.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

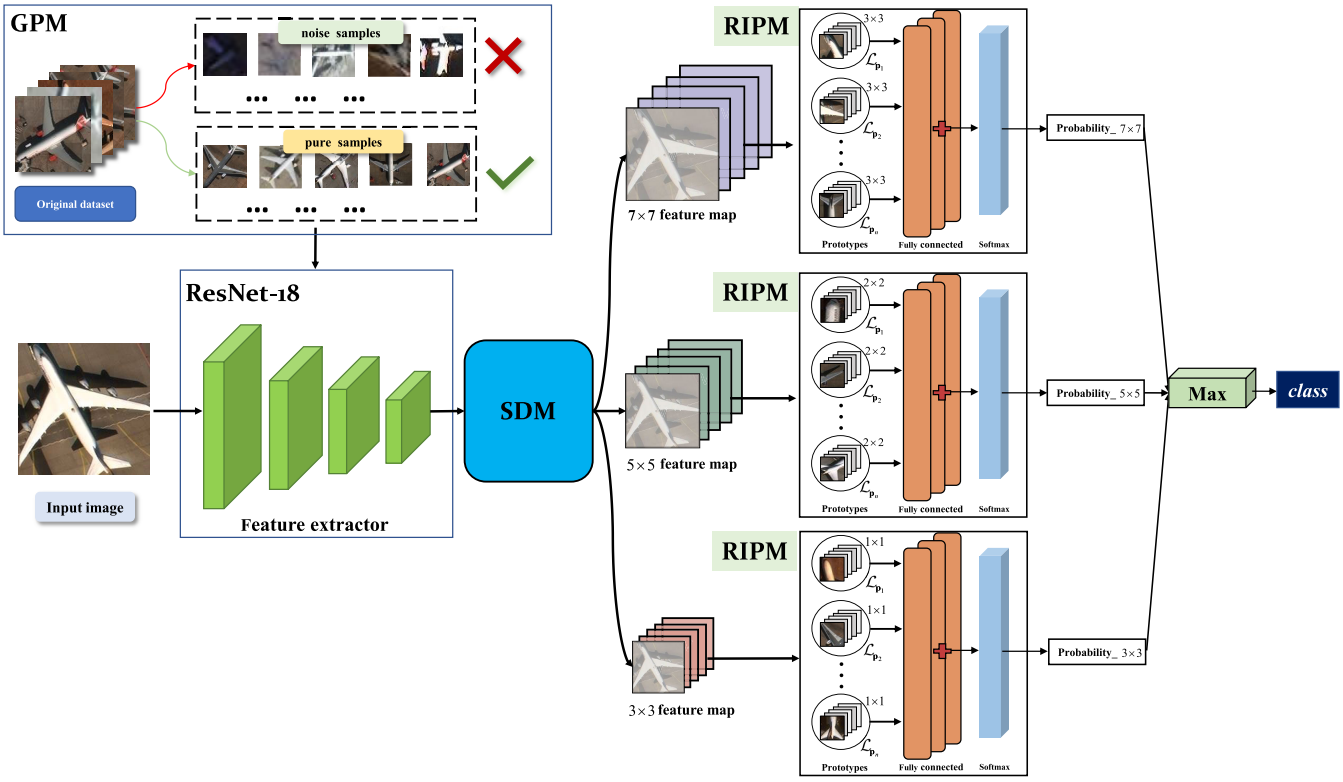


Fig. 3. Overall architecture of our network.

2) Noise Samples Selection: The essential component for identifying noise samples is the witness function [45], providing insights into the disparity between density estimates at a specific point. It can be computed as follows:

$$\text{witness}(x) = \frac{1}{n}\sum_{i=1}^{n} k_{\text{gauss}}(x, v_i) - \frac{1}{m}\sum_{j=1}^{m} k_{\text{gauss}}\left(x, p_j\right) \quad (5)$$

where the first term in the formula signifies the average proximity between sample x and the data, and the second term represents the average proximity between sample x and the prototypes.

If the witness function for a sample x is close to 0, it indicates that the density function of the data and the prototypes are closely aligned, implying that the distribution of prototypes resembles the distribution of the data at sample x. Finally, by calculating the absolute value of the witness function, the first c points with higher absolute values determined are the noise samples, this step is implemented through greedy search [46].

B. Rotation-Invariant Prototype Module

We utilized the prototypical part network (ProtoPNet) established by Chen et al. [25], including the feature extractor, prototype generation layer, and fully connected layer.

This network adopts a part-centric approach, learning and obtaining a template for each part of the target (component). Ultimately, through training, multiple templates are acquired and fused into a prototype set for the current category. During classification testing, each part of the test sample is compared separately with the prototype set of each class. The overall similarity of each part is synthesized, and prediction is made based on the weighted combination of the optimal class, ultimately determining the classification result.

In the RIPM, prototypes are allocated based on the aircraft's categories $\{1, \ldots, C\}$, with each category assigned k prototypes (here, $k = 5$). This choice is informed by considering each aircraft as composed of five distinct parts, hence deeming five prototypes sufficient to capture various components of the aircraft. RPIM will eventually learn a set of prototypes, which are intended to capture the most relevant components for identifying images of class c.

Specifically, each RIPM comprises n prototype layers denoted by $\{\mathcal{L}_{\mathbf{p}_1}, \ldots, \mathcal{L}_{\mathbf{p}_n}\}$, which is responsible for computing the distance between the prototype $\mathbf{p}_i$ and each patch in the convolutional network's output features. The calculated distances are then transformed into similarity scores using the logarithmic function, as defined by the following formula:

$$\mathcal{L}_{\mathbf{p}_i}(\mathbf{x}) = \max_{\tilde{\mathbf{x}} \in \text{patches}(\mathbf{x})} \log\left(-\text{cos_sim}(\tilde{\mathbf{x}}, \mathbf{p}_i) + 1\right). \quad (6)$$

Here, cosine similarity is employed to enhance the model's rotational invariance concerning aircraft targets. The objective is to ensure that aircraft targets in input images generate similar feature representations under different rotation angles. The computation of cosine similarity is expressed as follows:

$$\cos_\ \text{sim}\left(\tilde{\mathbf{x}}, \mathbf{p}_i\right) = \frac{\tilde{\mathbf{x}} \cdot \mathbf{p}_i}{\|\tilde{\mathbf{x}}\| \cdot \|\mathbf{p}_i\|}. \quad (7)$$

As cosine similarity approaches 1 for more similar vectors, $\mathcal{L}_{\mathbf{p}_i}$ is monotonically increasing relative to cos_sim.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

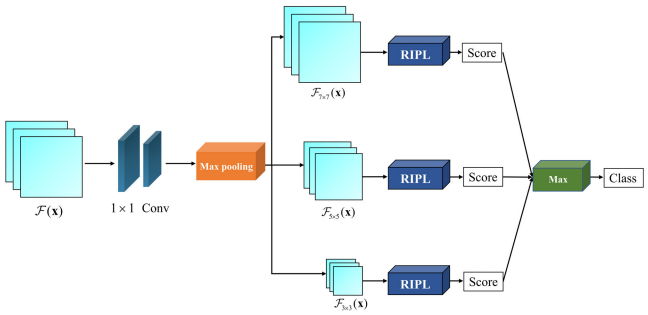


Fig. 4. Structure of the SDM.

Consequently, $\mathcal{L}_{\mathbf{p}_i}$ computes the similarity score for the patch in the image that is most similar to the ith rotation invariant prototype $\mathbf{p}_i$.

Ultimately, the n prototype layers $\mathcal{L}_{\mathbf{p}}$ yield n similarity scores, collectively assessing the similarity of each part of the input image to each prototype. These similarity scores are then multiplied by the weights in the fully connected layer [47] to output the prediction score for each category. Also, these prediction scores are used to predict probabilities through a softmax function, with the highest probability being chosen as the predicted class. The weight of the input score represents the importance of each prototype. Our method ensures that the prototype-based comparison is not affected by target rotation, enhancing the robustness of the network.

C. Scale Decision Module

Inspired by the spatial pyramid pooling [48], our SDM employs pooling layers to categorize features into distinct scales. This strategic scale segmentation aims to facilitate the learning of prototype components at different scales during the training phase and to judiciously match the correct prototype with the target feature scale during inference. The structure of this module is shown in Fig. 4.

First, we utilize 1×1 convolutional layers to reduce the dimensionality of input features, thereby mitigating parameter complexity and enhancing computational efficiency. The resulting dimensionally reduced feature is denoted by $\mathcal{F}(\mathbf{x})$. Also, we employ pooling layers to independently downsample the feature $\mathcal{F}(\mathbf{x})$ into three distinct scale-specific feature maps: $\mathcal{F}_{7\times7}(\mathbf{x})$, $\mathcal{F}_{5\times5}(\mathbf{x})$, and $\mathcal{F}_{3\times3}(\mathbf{x})$. Then, applying the RIPL independently at each scale to generate scores and determine the maximum score among the three scales.

Worth noting that the size of the prototypes at each scale is not uniform. Specifically, within the 7×7 scale, the prototype size in our prototype layer is set to 3×3. Transitioning to the 5×5 scale, the prototype size in our prototype layer is adjusted to 2×2. Within the 3×3 scale, the prototype size further reduces to 1×1 in our prototype layer. In the fine-grained identification of remote sensing aircraft targets, the scale and resolution of the target may vary greatly, and the adaptability of the model to different spatial characteristics is crucial. Our SDM enables different prototype sizes to enable the model to effectively capture complex details in high-resolution images while identifying broader contextual patterns, thereby helping the model to enhance its performance in identifying fine-grained features of aircraft targets at different scales.

D. Loss Function and Training Strategy

In this section, we will introduce the loss function and the training strategy of the proposed method.

1) Loss Function: Our MSRIP-Net mainly has three tasks. First, to make image patches in different scale feature spaces cluster around semantically similar prototypes while ensuring that they are not affected by rotation, the loss is defined as follows:

$$L_1 = \frac{1}{N}\sum_{i=1}^{N} \min_{j:\mathbf{p}_j \in \mathbf{P}_c} \min_{\tilde{\mathbf{x}} \in \text{patches}(\mathcal{F}(\mathbf{x}_i))} (1 - \cos_\text{sim}(\tilde{\mathbf{x}}, \mathbf{p}_j)) \quad (8)$$

where $\mathbf{P}_c = \{\mathbf{p}_j\}_j^k = 1$ represents the subset of prototypes assigned to category c. $\min_{\tilde{\mathbf{x}} \in \text{patches}(\mathcal{F}(\mathbf{x}_i))}$ selects the image patch $\tilde{\mathbf{x}}$ from the set of patches extracted by the feature extractor $\mathcal{F}$ for image i. Also, $\min_{\tilde{\mathbf{x}} \in \text{patches}(\mathcal{F}(\mathbf{x}_i))}$ selects the prototype $\mathbf{p}_j$ belonging to class c of the true label for image $\mathbf{x}_i$. The overall formulation involves averaging this loss across all training images, ensuring a balanced consideration across the entire training dataset.

This loss aims to minimize the complement of the cosine similarity between image patches and prototypes associated with their true class, encouraging the representations of training images in the latent space to be closer to prototypes of their respective classes [33], [49]. Utilizing cosine similarity introduces the invariance to rotations, enhancing the model's robustness to such transformations.

Second, to disentangle clusters centered around prototypes belonging to different categories, ensuring semantic distinctiveness, the loss is defined as follows:

$$L_2 = -\frac{1}{N}\sum_{i=1}^{N} \min_{j:\mathbf{p}_j \notin \mathbf{P}_c} \min_{\tilde{\mathbf{x}} \in \text{patches}(\mathcal{F}(\mathbf{x}_i))} (1 - \cos_\text{sim}(\tilde{\mathbf{x}}, \mathbf{p}_j)). \quad (9)$$

This loss aims to minimize the complement of the cosine similarity between image blocks and prototypes not associated with their true class [25]. This encourages the representations of training images in the latent space to be further away from prototypes not belonging to their true class, facilitating the formation of distinct clusters, each focusing on its respective category.

Third, to penalize the misclassifications on the training data, we use the cross-entropy loss [50], denoted as follows:

$$L_3 = \frac{1}{N}\sum_{i=1}^{N} \text{CrossEntropy}(\text{Net}(\mathbf{x}_i), y_i) \quad (10)$$

where $\text{Net}(\mathbf{x}_i)$ is the output of the model with input image $\mathbf{x}_i$.

This loss quantifies the dissonance between the predicted probability distribution and the actual label. Throughout the training, it steers model parameter updates by scrutinizing the forecast probability distribution in relation to the factual label distribution.

The ultimate optimization problem we aim to address encompasses the combination of the three aforementioned loss

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

functions

$$\text{Loss} = \lambda_1 L_1 + \lambda_2 L_2 + L_3 \tag{11}$$

where λ_1 and λ_2 are the two hyperparameters. For the coefficients of the cluster cost λ_1 and the separation cost λ_2, we considered three different settings: (1, 0.1), (0.8, 0.08), and (0.6, 0.06), and selected the pair (0.8, 0.08) using cross validation.

2) Training Strategy: In the first stage, we only update the weights in the feature extractor, SDM and RIPM before the fully connected layers by stochastic gradient descent (SGD) [51], aiming to learn the latent space at different scales. In this space, the L_1 loss is used to classify the training image into potential image patches that are close to at least one prototype of its true category, and the L_2 loss is used to classify each potential image patch away from prototypes that are not its true category. Also, the cross-entropy loss penalizes the misclassification of training data.

In order to make the model's inference process transparent, we adopt the approach from ProtoPNet [25] to project prototypes onto the training images. Specifically, for a prototype $\mathbf{p}_j$ belonging to class k, we use an argument function for projection

$$\mathbf{p}_j \leftarrow \arg\min_{\mathbf{z} \in \mathcal{Z}_j} \left\| \mathbf{z} - \mathbf{p}_j \right\|_2^2 \tag{12}$$

where $\mathcal{Z}_j = \left\{ \tilde{\mathbf{z}} : \tilde{\mathbf{z}} \in \text{patches}(\mathcal{F}(\mathbf{x}_i)) \forall i \text{ s.t. } y_i = k \right\}_{\circ}$. Here, we still use Euclidean distance, and the image block sizes at the three scales are 3×3, 2×2, and 1×1, respectively. The last stage is to optimize the classification weight layer.

IV. Experiments and Results

In this section, we will conduct extensive experiments to verify the effectiveness of MSRIP-Net. We elaborate the results on two datasets RarePlanes [52] and FAIR1M1.0 [53] separately, including ablation experiments, comparative experiments, and interpretable analyses.

Model training and evaluation were conducted on a Linux server equipped with the PyTorch framework, leveraging an RTX 3090 GPU. The training process comprised 200 epochs for the RarePlanes [52] dataset and 140 epochs for the FAIR1M1.0 [53] dataset.

A. Dataset Descriptions

RarePlanes [52] is an open-source dataset focused on the automatic detection of aircraft and their attributes in remote sensing images. The dataset comprises 22 475 instances of aircraft, categorized into seven classes: 1) small civil transport (SCT); 2) medium civil transport (MCT); 3) large civil transport (LCT); 4) military fighter (MF); 5) military bomber (MB); 6) military transport (MT); and 7) military trainer (MTR), the samples are shown in Fig. 5. The dataset is divided into civil and military categories based on roles. Each role is further classified into three sizes: large, medium, and small, effectively capturing the multiscale character of remote sensing images, which is one of the reasons we chose this dataset to validate our approach. In the experiment, we randomly selected 75% of each category as the training set and 25% as the test set. Table I shows the number of samples in each class, which can be clearly seen that the sample distribution of this dataset is extremely unbalanced. This poses a significant challenge for model training and evaluation, particularly concerning categories with less samples.

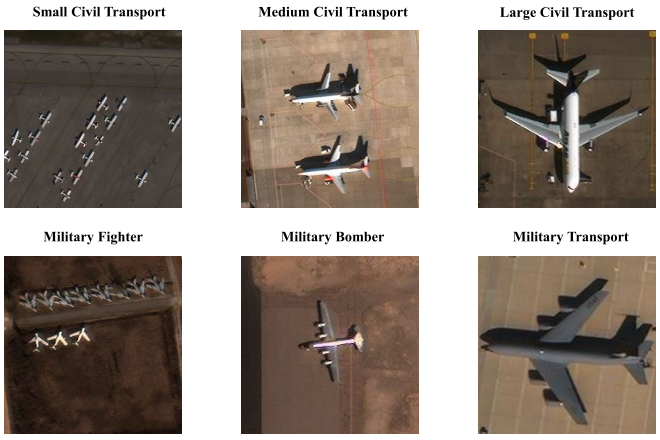


Fig. 5. RarePlanes data samples, (top) civil and (bottom) military.



Fig. 6. FAIR1M1.0 data samples.

TABLE I
Number of Training and Testing Samples for Each Class on RarePlanes

Class	LCT	MCT	SCT	MF	MTR	MT	MB	Total
Training	1008	5440	9779	173	17	253	6	**16676**
Testing	661	2379	2472	97	9	177	4	**5779**

FAIR1M1.0 [53] is a benchmark dataset for fine-grained object recognition in high-resolution remote sensing images. This dataset contains five types of objects: aircraft, ships, vehicles, courts, and roads, and we selected the aircraft targets among them for experiments. FAIR1M1.0 [53] contains ten classes of aircraft, i.e., Boeing 737 (B737), Boeing 747 (B747), Boeing 777 (B777), Boeing 787 (B787), Airbus A320 (A321), Airbus A220 (A220), Airbus A330 (A330), Airbus A350 (A350), COMAC C919 (C919), and COMAC ARJ21 (ARJ21), and the samples are shown in Fig. 6. We divide the dataset according to 4:1 for training and testing, and the number of samples in each class is shown in Table II. This dataset comprises subcategories within larger categories, and potential subtle yet crucial differences may exist among these

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

TABLE II
NUMBER OF TRAINING AND TESTING SAMPLES FOR EACH CLASS ON FAIR1M1.0 DATASET

Class	B737	B747	B777	B787	A220	A321	A330	A350	ARJ21	C919	Total
Training	3949	1673	1532	1669	6056	2504	1599	1064	166	135	**20347**
Testing	987	418	383	425	1514	626	400	266	42	34	**5095**

TABLE III
DIFFERENT BACKBONES PERFORMANCES ON THE RAREPLANES DATASET WITH THE BEST RESULTS HIGHLIGHTS IN BOLD

Backbone	SCT	MCT	LCT	MF	MB	MT	MTR	OA(%)	KAPPA
ResNet-18	**95.37**	89.73	**95.58**	89.69	**75.00**	94.92	88.89	**92.97**	**0.625**
ResNet-34	92.84	90.72	90.09	90.87	75.00	90.13	87.29	88.28	0.603
ResNet-50	94.17	**91.33**	86.24	85.07	75.00	**95.25**	**89.95**	88.15	0.607
VGG-16	90.61	86.21	82.36	**91.65**	50.00	89.14	89.31	82.75	0.583
VGG-19	88.25	89.02	85.45	86.20	50.00	90.08	84.12	81.88	0.576
MobileNetV3	86.38	82.35	86.56	86.42	50.00	87.43	86.50	80.81	0.532
ShuffleNetV2	87.04	84.12	82.13	83.15	25.00	86.37	83.19	74.71	0.483
GhostNet	89.57	87.93	81.30	80.74	50.00	84.38	79.21	77.48	0.502
ViT	77.95	80.19	74.86	82.10	25.00	73.33	83.10	70.40	0.461
Swin	80.37	81.71	82.64	77.32	50.00	80.52	84.91	75.52	0.487

classes. Consequently, it is highly suitable for validating the effectiveness of fine-grained recognition methods.

B. Backbone Selection of the Method

In this section, we will present the performance of our method under different backbone networks to justify the final selection of ResNet-18 [44]. Specifically, we employed ResNet-18 [44], ResNet-34 [44], ResNet-50 [44], VGG-16 [54], VGG-19 [54], three lightweight networks: MobileNetV3 [55], ShuffleNetV2 [56], and GhostNet [57], as well as two Transformer architecture networks: ViT [58] and Swin [59], respectively, as the backbone for feature extraction in our method.

Tables III and IV present the top-1 accuracy per class, overall accuracy (OA), and kappa coefficient using different backbones on two datasets. It can be observed that the ResNet-18 backbone performs relatively well overall, although its accuracy on certain classes may be lower than that of ResNet-34 and ResNet-50. This can be attributed to the superior performance of prototype selection in the shallow feature space of images. In addition, considering the deployment of the model at the edge, both parameter count and computational complexity are factors that need to be taken into account. Therefore, backbone architectures based on Transformer were excluded from consideration. Taking all factors into consideration, we select ResNet-18 as the backbone for our method.

C. Results on Rareplanes

1) Visualization of Reasoning Process: In this section, we will present the reasoning process of MSRIP-Net for classifying aircraft targets in the test set, including the similarity scores obtained for each part of the network with the selected prototypes. In addition, we will provide information on which training set image the prototype is derived from and showcase this information by overlaying it on the original image in the form of a heatmap. Specifically, for each aircraft class c, MSRIP-Net is capable of providing evidence for belonging to the category with the highest probability score, ensuring a high level of interpretability.

For example, Fig. 7 illustrates the reasoning process for determining that a test aircraft target belongs is "LCT" rather than "MT." By calculating the similarity score between the nose part of the test image and the learned LCT nose part prototype, our network knows that the target's nose has a high similarity with LCT. Similarly, the same approach was taken for the left and right wings and tail parts. These similarity scores are weighted and summed together to obtain the final score e for the aircraft belonging to this class. It is obvious that the score of LCT is much higher than that of MT, so finally, the target is classified as LCT. And from the figure, even if the target undergoes rotation, our MSRIP-Net can still effectively match the similarity between various parts of the target and their corresponding prototypes.

2) Comparisons With the SOTA Methods: In order to better highlight the effectiveness of our method, we compare it with different models, which can be divided into three categories: 1) traditional CNN methods, such as ResNet-18 [44], VGG-19 [54], and BCNN [60]; 2) classification methods based on attention mechanism, such as EFM-Net [21], SFRNet [61], Adap-Net [9], and MIL [62]; and 3) classification method based on aircraft part prior knowledge, such as APPEAR [37].

In our experiments, the OA and KAPPA coefficient are used for evaluating the model's performance. In OA evaluation, we further utilize top-1 and top-5 accuracy to deeply analyze

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

TABLE IV
DIFFERENT BACKBONES PERFORMANCES ON THE FAIR1M1.0 DATASET WITH THE BEST RESULTS HIGHLIGHTS IN BOLD

Backbone	B737	B747	B777	B787	A220	A321	A330	A350	ARJ21	C919	OA(%)	KAPPA
ResNet-18	53.40	**96.41**	61.88	**81.41**	71.53	85.62	**92.25**	**90.60**	**73.81**	**79.41**	**74.58**	**0.631**
ResNet-34	52.17	92.74	59.29	79.30	**73.46**	81.74	88.57	89.91	71.25	76.51	73.04	0.615
ResNet-50	51.37	93.98	**62.54**	78.32	68.52	**85.74**	89.41	87.24	70.55	76.20	72.87	0.578
VGG-16	**55.31**	90.41	57.10	76.13	68.98	83.61	89.20	88.07	71.30	74.28	70.72	0.537
VGG-19	50.16	93.23	57.74	73.92	68.26	81.73	89.03	85.24	66.51	74.38	68.11	0.553
MobileNetV3	47.85	93.15	57.93	77.13	68.56	78.85	84.43	85.39	69.07	74.26	63.17	0.521
ShuffleNetV2	44.58	91.21	52.74	73.69	66.43	82.32	84.61	86.37	68.36	75.93	63.25	0.461
GhostNet	46.23	87.17	52.73	77.59	69.08	76.31	83.47	82.17	70.26	73.58	64.65	0.487
ViT	43.18	86.35	52.75	70.29	66.01	76.22	85.36	81.97	64.51	71.36	60.29	0.430
Swin	45.28	88.16	54.73	72.82	65.09	76.47	86.83	83.29	69.56	73.18	62.10	0.451

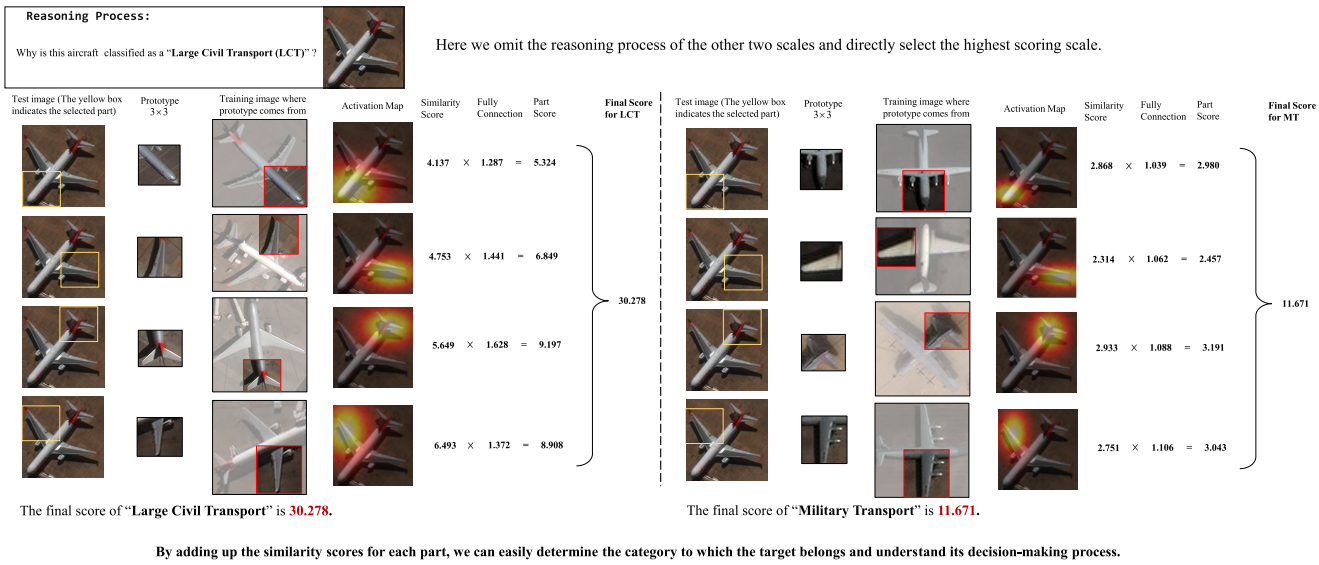


Fig. 7. Reasoning process on Rareplanes.

the performance of the model. OA is a measure of the overall performance of a model on the entire test set, where the top-1 accuracy focuses on the model's correct prediction ratio on the single most likely category, while the top-5 accuracy evaluates the model's performance in fuzzy and multiclassification task scenarios.

As shown in Table V, our method stands out as the top performer in both top-1 and top-5 evaluations, underscoring its exceptional capabilities in fine-grained classification tasks. In the top-1 accuracy assessment, our model consistently outperforms competitors, showcasing its ability to make precise and accurate predictions for each sample. This shows that our method can well capture complex details and identify subtle differences through data processing and prototype matching. Moreover, the superiority observed in the top-5 accuracy evaluation further accentuates the robustness of our method in multiclassification tasks. Surpassing not only traditional CNN methods but also classification methods based on attention mechanisms, our method demonstrates a comprehensive understanding and consideration of a broader range of potential predictions.

TABLE V
EXPERIMENTAL RESULTS OF OA AND KAPPA COEFFICIENT ON THE RAREPLANES DATASET

Method	Backbone	OA(%)		KAPPA
		top-1 accuracy	top-5 accuracy	
VGG-19	ResNet-18	74.09	80.16	0.436
ResNet-18	ResNet-18	79.19	83.57	0.451
BCNN	ResNet-18	83.35	86.28	0.457
MIL	ResNet-18	84.22	87.04	0.453
Adap-Net	ResNet-18	84.32	87.65	0.471
EFM-Net	ResNet-18	87.59	91.76	0.468
SFRNet	ResNet-18	86.56	93.39	0.503
APPEAR	ResNet-18	89.33	96.21	0.505
MSRIP-Net	ResNet-18	**92.97**	**99.78**	**0.625**

Furthermore, we emphasize that the KAPPA coefficient is consistently higher for our method compared with the three categories of methods mentioned above. This highlights the superior consistency and reliability of our model's predictions, substantiating its effectiveness in fine-grained classification

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
EXPERIMENTAL RESULTS OF TOP-1 ACCURACY FOR EACH CATEGORY ON THE RAREPLANES DATASET

Method	VGG-19	ResNet-18	BCNN	MIL	Adap-Net	EFM-Net	SFRNet	APPEAR	MSRIP-Net
SCT	83.8	86.32	89.84	88.9	87.99	92.96	91.84	93.04	**95.37**
MCT	65.15	72.7	79.09	78.87	79.44	80.26	82.42	86.02	**89.73**
LCT	78.22	86.11	82.54	92.67	93.47	95.25	88.10	92.72	**95.58**
MF	47.42	52.58	62.89	61.86	67.01	79.38	59.79	63.92	**89.69**
MB	0	0	25.00	0	0	50.00	50.00	50.00	**75.00**
MT	59.89	63.84	70.06	76.84	78.53	88.14	83.05	86.44	**94.92**
MTR	11.11	11.11	22.22	33.33	56.56	66.67	33.33	66.67	**88.89**

tasks. The higher KAPPA coefficient underlines the model's ability to achieve agreement with ground-truth annotations, surpassing not only traditional CNN methods but also models based on attention mechanisms and aircraft part prior knowledge.

In addition, we performed a top-1 accuracy evaluation for each category. As shown in Table VI, although the OA of the attention mechanism-based methods is higher than that of traditional CNN, the classification accuracy of less sample categories is still very low. It is particularly noteworthy that in the RarePlanes dataset, due to the extremely imbalanced distribution of training data, for example, the number of civilian aircraft samples is 36 times that of military targets. Despite this, our approach maintained excellent accuracy in each specific category, outperforming CNN-based and attention-based methods. This is because our method does not rely on a large amount of sample data for learning, but rather identifies representative partial prototypes in each type of sample and uses these prototypes to capture key partial features in test samples for comparative classification. This prototype-based matching approach enables our method to better adapt to imbalanced data distributions and scenarios with limited samples, as it focuses on the crucial parts within the samples rather than relying on the entire dataset.

3) Analysis of the Confusion Matrices: In order to further analyze the effectiveness of the proposed method, we conducted experiments comparing MSRIP-Net with the suboptimal method APPEAR [37], and the resulting confusion matrices are presented in Fig. 8. APPEAR exhibited challenges in distinguishing between the MF and SCT categories. The observed difficulty indicates a limitation in APPEAR's ability to capture the nuanced differences between these two classes. In addition, APPEAR struggled with aircraft at different scales within the civil transport category, highlighting a deficiency in its capacity to handle multiscale information effectively. This inadequacy is manifested in its frequent confusion between aircraft of varying sizes. In contrast, our MSRIP-Net shows superior performance in capturing subtle differences between military and civilian aircraft. Moreover, it excels in accurately matching aircraft of diverse scales. This accomplishment underscores the robustness of our approach in processing multiscale information and distinguishing between visually similar categories, surpassing the limitations observed in the comparative method APPEAR.

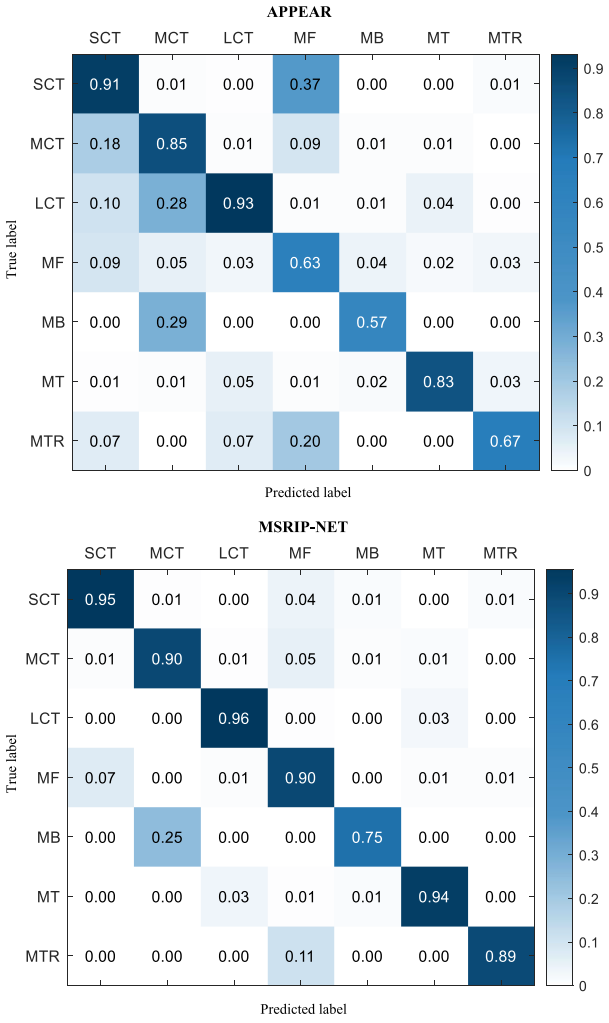


Fig. 8. Confusion matrices for APPEAR and the proposed MSRIP-NET on the RarePlanes dataset.

In conclusion, the in-depth analysis of confusion matrices reinforces the effectiveness of MSRIP-Net in fine-grained aircraft recognition, showcasing its superiority over the suboptimal method APPEAR in capturing intricate category differences and handling multiscale variations with finesse.

4) Ablative Analysis: To systematically evaluate the effectiveness of the proposed MSRIP-Net and understand the contribution of its individual components, we conduct ablation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

TABLE VII
ABLATION EXPERIMENTAL RESULTS ON THE RAREPLANES DATASET

No.	Components				Top-1 Accuracy							OA(%)
	ResNet-18	GPM	RIPM	SDM	SCT	MCT	LCT	MF	MB	MT	MTR	
1	✓				86.32	72.70	86.11	52.58	0.00	63.84	11.11	79.19
2	✓	✓			89.60	79.35	87.15	61.28	0.00	70.26	11.11	82.05
3	✓		✓		91.28	86.91	89.47	84.32	**75.00**	88.64	**88.89**	84.73
4	✓	✓	✓		92.1	87.62	91.26	87.35	**75.00**	91.85	**88.89**	85.88
5	✓	✓	✓	✓	**95.34**	**89.43**	**95.58**	**89.69**	**75.00**	**94.92**	**88.89**	**92.97**

experiments by selectively combining different modules. There are five groups of experiments, and the results are shown in Table VII.

Through the comparison of experiments 1 and 2, we observe a positive impact of our proposed GPM on the model's performance. The results indicate that the GPM module, built upon the baseline ResNet-18, enhances the model's focus on genuine aircraft samples by filtering out noise samples, thereby improving both per-class accuracy and overall performance. However, the effectiveness of the GPM module is contingent upon a sufficiently abundant dataset. It is evident that the inclusion of GPM does not yield significant improvements for classes with extremely limited samples.

On another note, by contrasting experiment 1 with experiment 3, the introduction of the RIPM significantly improves each category's accuracy, especially in categories prone to confusion under the same scale. It is worth noting that through Experiments 1–5, we can find that for the extremely limited sample categories MF and MTR, the accuracy rate is significantly improved when RIPM is added. This suggests that RIPM effectively addresses issues related to the model's performance variation under changes in orientation, posture, and position, enabling accurate recognition of aircraft across different orientations and postures.

Further comparisons between experiment 4 and experiment 5 demonstrate the effectiveness of the SDM. The results reveal a noticeable enhancement in category accuracy, particularly for classes at different scales (e.g., LCT and MCT), showcasing the successful resolution of multiscale matching challenges by the SDM module. This ensures the model's improved adaptability to target recognition tasks across various scales. In summary, the stepwise introduction of GPM, RIPM, and SDM modules yields significant improvements in each category's accuracy and overall performance. This substantiates the effectiveness of our proposed modules in addressing issues related to noise, orientation invariance, and multiscale matching, providing a comprehensive solution for fine-grained aircraft recognition tasks.

5) Visualization Results on Rareplanes: The visualization results are shown in Fig. 9, where the red boxes indicate incorrectly identified results. It is observed that our method demonstrates outstanding performance in recognizing different categories, regardless of scale or categories with minimal samples. This further substantiates the superior performance of our method in handling noisy samples, rotating objects, and multiple scales.

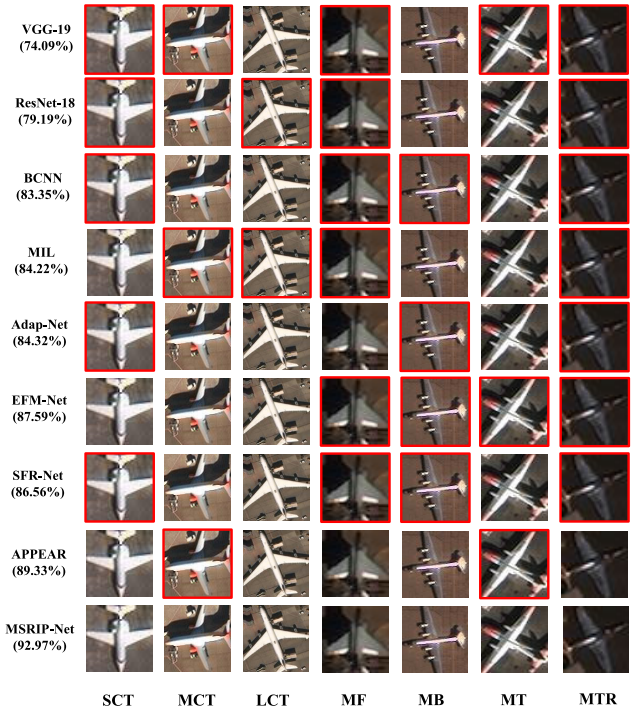


Fig. 9. Visualization results of aircraft recognition with the Rareplanes dataset.

D. Results on FAIR1M1.0

1) Visualization of Reasoning Process: In this section, we provide a detailed showcase of the decision-making process employed by the MSRIP-Net model, specifically in identifying the target as a Boeing 747 rather than a Boeing 777, as illustrated in Fig. 10. Due to the high similarity among the targets in the FAIR1M1.0 [53] dataset, belonging to subcategories within a broader category, an issue arises during the computation of scores for individual components. This poses a challenge during the computation of scores for individual components, leading to scores for the incorrect category surpassing those for the correct category. For instance, in the third row of Fig. 10, the tail of the Boeing 747 is even more similar to the prototype of our learned Boeing 777 tail, surpassing the score of the Boeing 747 tail prototype. Simultaneously, their noses bear a striking resemblance. However, there is a notable discrepancy in the similarity of their left and right wings, and as such, our model successfully captures this distinguishing feature, revealing a significant difference in the similarity

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

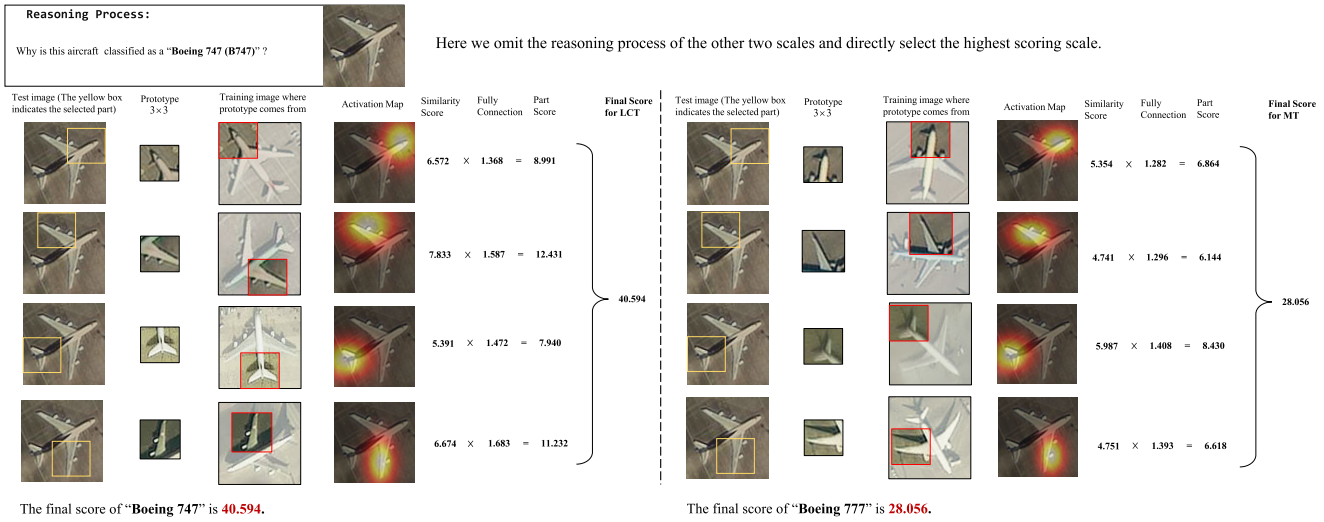


Fig. 10. Reasoning process on FAIR1M1.0.

TABLE VIII
EXPERIMENTAL RESULTS OF OA AND KAPPA COEFFICIENT ON THE FAIR1M1.0 DATASET

Method	Backbone	OA(%)		KAPPA
		top-1 accuracy	top-5 accuracy	
VGG-19	ResNet-18	48.20	53.76	0.421
ResNet-18	ResNet-18	55.49	58.16	0.471
BCNN	ResNet-18	58.41	62.85	0.523
MIL	ResNet-18	61.18	63.09	0.562
Adap-Net	ResNet-18	60.75	64.88	0.558
EFM-Net	ResNet-18	61.57	65.23	0.577
SFRNet	ResNet-18	63.75	68.92	0.584
APPEAR	ResNet-18	68.38	73.84	0.609
MSRIP-Net	ResNet-18	**74.58**	**80.60**	**0.631**

scores of the left and right wings. Ultimately, by adding up the similarity scores of each part, our model accurately classifies it as a Boeing 747.

2) Comparisons With the SOTAMethods: In this section, we will prove the effectiveness of our method by comparing it with the SOTA methods. The specific comparison methods and evaluation indicators used are the same as that shown in Section IV-C2. As shown in Table VIII, traditional deep learning methods exhibit relatively mediocre OA in this multiclassification task, with OAs of 48.20%, 55.49%, and 58.41%, respectively. This can be attributed to their challenges in effectively capturing subtle differences among different aircraft targets, especially when dealing with complex fine-grained classification problems. In comparison, methods based on attention mechanisms (MIL, Adap-Net, EFM-Net, and SFRNet) achieve higher OAs, specifically 61.18%, 60.75%, 61.57%, and 63.75%. This suggests that the introduction of attention mechanisms allows models to focus and emphasize key features, thereby improving overall classification performance. However, despite the significant improvement in OA, these methods do not exhibit a notable enhancement in top-1 accuracy compared with traditional CNN methods. This may indicate that there is still room for improvement in their accuracy across all categories. It is noteworthy that our proposed prototype-based interpretable method demonstrates a significant advantage in accuracy. By calculating similarity scores for each part of the aircraft based on categories, this method accurately captures minute feature differences, resulting in substantial improvements in top-1 and top-5 OAs. Moreover, the comprehensive evaluation provided by the KAPPA coefficient offers a more thorough reflection of the model's performance in multiclassification tasks.

As shown in Table IX, MSRIP-Net achieves the highest top-1 accuracy across all categories. Particularly, MSRIP-Net performs exceptionally well in the B737 and A330 categories, achieving accuracy rates of 96.41% and 92.25%, respectively. This demonstrates the superiority of MSRIP-Net in accurately identifying small categories in fine-grained targets. Moreover, the recognition accuracy for the small-sample categories ARJ21 and C919 in the dataset also reaches 73.81% and 79.41%, respectively, significantly outperforming other methods.

3) Analysis of the Confusion Matrices: To further establish the superior performance of our method in fine-grained recognition, a comprehensive comparison was conducted against the SOTA attention-based method, SFRNet. The evaluation utilized a confusion matrix, as illustrated in Fig. 11.

In the examination of Boeing series aircraft, SFRNet encountered challenges in handling subtle variations, resulting in considerable confusion among the B737, B747, and B777 classes. Furthermore, its classification accuracy for A220 was suboptimal. Particularly, noteworthy were the difficulties faced by SFRNet in distinguishing between two independent small-sample classes, ARJ21 and C919, often misclassifying them as B737.

In contrast, our proposed method, MSRIP-Net, demonstrated remarkable performance in addressing these challenges.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

TABLE IX
EXPERIMENTAL RESULTS OF TOP-1 ACCURACY FOR EACH CATEGORY ON THE FAIR1M1.0 DATASET

Method	VGG-19	ResNet-18	BCNN	MIL	Adap-Net	EFM-Net	SFRNet	APPEAR	MSRIP-Net
B737	37.69	40.63	42.86	44.98	43.87	45.69	47.42	49.95	**53.40**
B747	68.42	73.44	77.51	79.67	76.32	80.62	81.37	90.43	**96.41**
B777	20.37	25.59	28.46	30.55	32.11	34.20	33.68	41.25	**61.88**
B787	54.35	59.53	62.59	65.41	64.01	61.41	66.59	72.71	**81.41**
A220	43.13	58.92	60.17	62.95	61.96	62.52	64.49	67.24	**71.53**
A321	65.81	69.17	73.00	75.59	78.12	74.76	76.89	80.35	**85.62**
A330	59.50	61.75	68.75	74.50	80.25	77.25	83.25	89.25	**92.25**
A350	62.78	64.29	68.42	71.80	74.43	78.19	75.93	82.71	**90.60**
ARJ21	23.80	30.95	35.71	40.48	38.10	33.34	42.85	61.90	**73.81**
C919	26.47	35.29	41.18	44.12	38.24	32.36	44.12	64.71	**79.41**

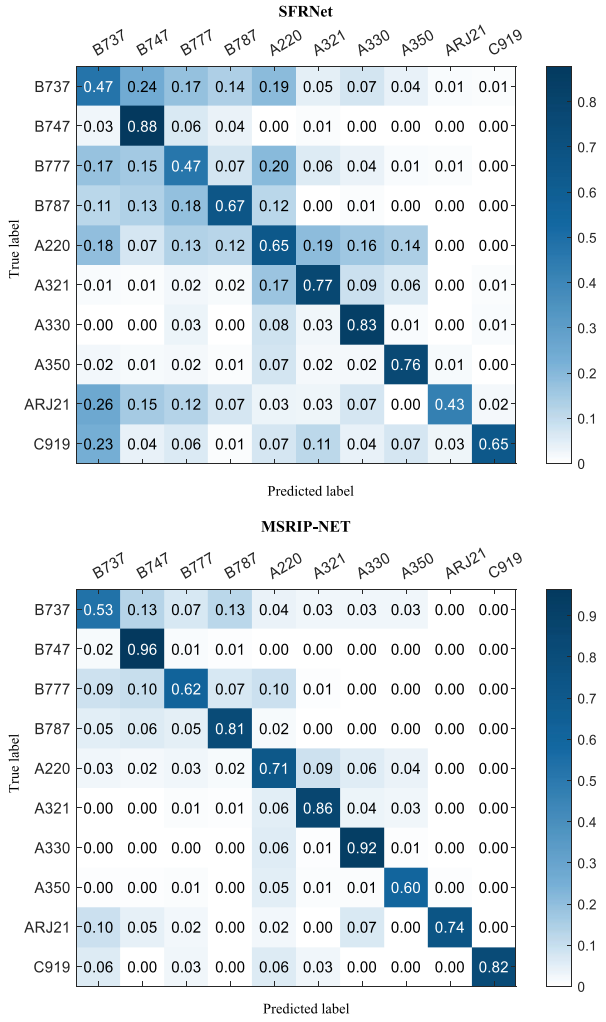


Fig. 11. Confusion matrices for SFRNet and the proposed MSRIP-NET on the FAIR1M1.0 dataset.

Our approach excelled in capturing subtle differences among highly similar targets, affirming its effectiveness in fine-grained recognition. Specifically, MSRIP-Net showcased precision in differentiating between Boeing series aircraft, overcoming the challenges encountered by SFRNet. Moreover, our method exhibited exceptional discriminative power for small-sample classes (ARJ21 and C919), accurately identifying them even in the presence of similarities with other classes.

These findings underscore the efficacy of MSRIP-Net in achieving robust and accurate fine-grained object recognition, positioning it as a compelling solution for challenging scenarios with intricate category differences and limited sample sizes.

4) Ablative Analysis: Table X presents the results of ablation experiments conducted on the FAIR1M1.0 [53] dataset to analyze the impact of various components in MSRIP-Net.

It can be seen from Experiments 1 and 2 that after adding GPM, the accuracy of each category is improved, which indicates that GPM effectively filters noisy samples, strengthening the model's attention toward real-scene aircraft scenarios. However, the efficacy of the GPM module relies on a sufficient quantity of data for precise filtering. Consequently, its effects are less pronounced in small-sample categories (ARJ21 and C919).

By comparing experiment 1 with experiment 3, where the RIPM was introduced, we can see precision significantly improves in easily confusable categories (B737, B747, B777, and B787). This enhancement signifies the efficacy of our core module, RIPM, in autonomously learning meaningful part prototypes, enabling precise discrimination of subtle differences. Furthermore, the accuracy gains extend to small-sample categories (ARJ21 and C919), indicating a substantial improvement in recognition performance for these challenging cases. This observation underscores the pivotal role of RIPM in fine-grained aircraft recognition. By autonomously identifying and understanding meaningful part prototypes, RIPM enhances the model's ability to discern nuanced differences, especially in categories where visual similarities pose challenges. In summary, the outcomes of Experiment 3 demonstrate the effectiveness of RIPM in addressing challenges related to confusable categories and small-sample classes, establishing it as a key component contributing to the overall success of MSRIP-Net in fine-grained object recognition.

Finally, we demonstrate the impact of the SDM through further comparisons in Experiments 4 and 5. The SDM module,

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

TABLE X
ABLATION EXPERIMENTAL RESULTS ON THE FARI1M1.0 DATASET

No.	Components				Top-1 Accuracy										OA(%)
	ResNet-18	GPM	RIPM	SDM	B737	B747	B777	B787	A220	A321	A330	A350	ARJ21	C919	
1	✓				40.62	73.44	25.59	59.53	58.92	69.17	61.75	64.29	30.95	35.29	55.49
2	✓	✓			44.83	80.14	31.82	66.85	62.13	73.58	72.68	69.74	35.69	39.63	60.15
3	✓		✓		48.37	87.96	53.27	73.20	66.41	79.38	80.36	78.43	60.18	68.29	67.31
4	✓	✓	✓		50.26	92.35	58.32	77.09	68.57	81.65	85.34	84.97	65.19	74.26	70.81
5	✓	✓	✓	✓	**53.39**	**96.41**	**61.88**	**81.41**	**71.53**	**85.62**	**92.25**	**90.60**	**73.81**	**79.41**	**74.58**

TABLE XI
COMPARISON OF MODEL PARAMETER COUNTS

Method	VGG-19	ResNet-18	BCNN	MIL	Adap-Net	EFM-Net	SFRNet	APPEAR	**MSRIP-Net**
Params (MB)	143.69	11.75	1843.19	56.91	12.43	91.76	56.48	47.62	**18.35**

TABLE XII
AVERAGE INFERENCE TIME PER SINGLE IMAGE COMPARISON OF DIFFERENT METHODS ACROSS VARIOUS BATCH SIZES

Batch Size	**Method**								
	VGG-19	ResNet-18	BCNN	MIL	Adap-Net	EFM-Net	SFRNet	APPEAR	**MSRIP-Net**
8	0.75s	0.42s	1.28s	0.57s	0.46s	0.61s	0.55s	1.05s	0.53s
16	0.83s	0.57s	1.31s	0.68s	0.51s	0.68s	0.62s	1.14s	0.65s
32	0.95s	0.63s	1.37s	0.79s	0.59s	0.83s	0.77s	1.31s	0.72s
64	1.08s	0.74s	1.65s	0.91s	0.68s	0.96s	0.95s	1.37s	0.87s

by matching prototypes across three scales, mitigates the challenges posed by multiscale variations in remote sensing images. For instance, in small-target categories such as ARJ21 and C919, where models might struggle due to similarities with larger Boeing series targets, SDM proves effective in preventing misclassifications.

5) Visualization Results on FAIR1M1.0: The visualization results are shown in Fig. 12, where the red boxes indicate incorrectly identified results. It is observed that a higher error rate for the ARJ21 category, which exhibits severe deformation and shading. For the four variants of the Boeing series aircraft (B737, B747, B777, and B787), their structural and shape similarities lead to confusion in identification for most methods. APPEAR and our method distinguish these aircraft well by considering them as rigid structures. However, APPEAR struggles with significant scale changes in the A321 category, further highlighting the superiority of our approach.



Fig. 12. Visualization results of aircraft recognition with the FAIR1M1.0 dataset.

E. Discussion on Parameter Count and Time Cost

In this section, we provide a comparative analysis and discussion on the parameter count and inference time of the models. As shown in Table XI, ResNet-18 exhibits the smallest parameter count, totaling only 11.75 MB. It is noteworthy that our method is built upon ResNet-18 as its backbone architecture, resulting in an additional 6.6 MB of parameters. These supplementary parameters primarily come from prototype selection and the final classification head. Due to the intricate network architecture, BCNN exhibits the highest parameter count, approximately 1.8 GB. VGG-19, due to its deeper network depth, entails a larger parameter count, approximately

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

143.69 MB. Conversely, Adap-Net, employing a lightweight design, features a smaller parameter count, around 12.43 MB. The remaining methods all fall within the range of 100 MB. In summary, our approach utilizes a lightweight backbone network, ResNet-18, resulting in a smaller parameter count compared with most methods. However, it is important to note that our current focus is not on lightweight operations; hence, it may not represent the optimal solution in this regard.

In addition, we conducted tests on the average inference time per single image of the models. In order to provide readers with a more comprehensive performance comparison, we tested different methods at varying batch sizes. As shown in Table XII, ResNet-18 exhibits the smallest inference time per single image across different batch sizes. Similarly, Adap-Net, leveraging lightweight operations, demonstrates notable efficiency, with an average inference time of only 0.56-s per single image. Conversely, methods with higher computational costs include BCNN and APPEAR. The former due to the drawbacks associated with its bilinear networks characteristics, and the latter is divided into two stages: segmentation and classification. In summary, our method involves prototype matching and scale matching operations during inference, so its computational cost is slightly higher than ResNet-18 but lower than most methods.

V. Conclusion

This article introduced a novel MSRIP-Net for fine-grained aircraft recognition in remote sensing images. The proposed network leverages a GPM, RIPM, and SDM to enhance its ability to recognize intricate details, handle noise, deformation, and multi-scale variations inherent in remote sensing aircraft targets.

Experimental results on the FAIR1M1.0 dataset and the Rareplane fine-grained dataset showcased the superiority of MSRIP-Net over SOTA methods. Our network achieved a new SOTA recognition performance, particularly excelling in discriminating between Boeing series aircraft and improving accuracy for small-sample classes (ARJ21 and C919). The decision-making process demonstrated the interpretability of our method, providing transparency in recognizing diverse aircraft categories. In future work, we aim to extend the applicability of MSRIP-Net to broader fine-grained recognition tasks and explore enhancements to further boost its interpretability and robustness.

References

[1] Q. Liu, X. Xiang, Y. Wang, Z. Luo, and F. Fang, "Aircraft detection in remote sensing image based on corner clustering and deep learning," *Eng. Appl. Artif. Intell.*, vol. 87, Jan. 2020, Art. no. 103333.

[2] T. Shi et al., "Complex optical remote-sensing aircraft detection dataset and benchmark," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5612309.

[3] A. Zhao et al., "Aircraft recognition based on landmark detection in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 14, no. 8, pp. 1413–1417, Aug. 2017.

[4] G. Liu, X. Sun, K. Fu, and H. Wang, "Aircraft recognition in high-resolution satellite images using Coarse-to-Fine shape prior," *IEEE Geosci. Remote Sens. Lett.*, vol. 10, no. 3, pp. 573–577, May 2013.

[5] F. Zhang, B. Du, L. Zhang, and M. Xu, "Weakly supervised learning based on coupled convolutional neural networks for aircraft detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 9, pp. 5553–5563, Sep. 2016.

[6] W. Diao, X. Sun, F. Dou, M. Yan, H. Wang, and K. Fu, "Object recognition in remote sensing images using sparse deep belief networks," *Remote Sens. Lett.*, vol. 6, no. 10, pp. 745–754, 2015.

[7] X. Li, B. Jiang, S. Wang, L. Shen, and Y. Fu, "A human–computer fusion framework for aircraft recognition in remote sensing images," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 2, pp. 297–301, Feb. 2020.

[8] H. Wang, Y. Gong, Y. Wang, L. Wang, and C. Pan, "DeepPlane: A unified deep model for aircraft detection and recognition in remote sensing images," *J. Appl. Remote Sens.*, vol. 11, no. 4, 2017, Art. no. 042606.

[9] Y. Nie, C. Bian, and L. Li, "Adap-EMD: Adaptive EMD for aircraft fine-grained classification in remote sensing," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, 2022, Art. no. 6509005.

[10] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," *Commun. ACM*, vol. 60, no. 6, pp. 84–90, May 2017.

[11] K. He, G. Gkioxari, P. Dollár, and R. Girshick, "Mask R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2980–2988.

[12] R. Girshick, "Fast R-CNN," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Dec. 2015, pp. 1440–1448.

[13] P. Wang and N. Vasconcelos, "A generalized explanation framework for visualization of deep learning model predictions," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 8, pp. 9265–9283, Aug. 2023.

[14] D. Chang et al., "Making a bird AI expert work for you and me," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 45, no. 10, pp. 12068–12084, Oct. 2023.

[15] B. Zhou, A. Khosla, A. Lapedriza, A. Oliva, and A. Torralba, "Learning deep features for discriminative localization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2921–2929.

[16] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 618–626.

[17] Z. Huang and Y. Li, "Interpretable and accurate fine-grained recognition via region grouping," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 8662–8672.

[18] J. Wagner, J. M. Köhler, T. Gindele, L. Hetzel, J. T. Wiedemer, and S. Behnke, "Interpretable and fine-grained visual explanations for convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 9097–9107.

[19] P. Shaw, J. Uszkoreit, and A. Vaswani, "Self-attention with relative position representations," 2018, *arXiv:1803.02155*.

[20] W. Xiong, Z. Xiong, and Y. Cui, "An explainable attention network for fine-grained ship classification using remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5620314.

[21] Y. Yi, Y. You, C. Li, and W. Zhou, "EFM-net: An essential feature mining network for target fine-grained classification in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5606416.

[22] Y. Han, X. Yang, T. Pu, and Z. Peng, "Fine-grained recognition for oriented ship against complex scenes in optical remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5612318.

[23] Y. Fu, Z. Liu, and Z. Zhang, "Progressive learning vision transformer for open set recognition of fine-grained objects in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5215113.

[24] C. Pan, R. Li, Q. Hu, C. Niu, W. Liu, and W. Lu, "Contrastive learning network based on causal attention for fine-grained ship classification in remote sensing scenarios," *Remote Sens.*, vol. 15, no. 13, p. 3393, Jul. 2023.

[25] C. Chen, O. Li, D. Tao, A. Barnett, C. Rudin, and J. K. Su, "This looks like that: Deep learning for interpretable image recognition," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 32, 2019, pp. 1–12.

[26] H. Bilen and A. Vedaldi, "Weakly supervised deep detection networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 2846–2854.

[27] K. Yang, D. Li, and Y. Dou, "Towards precise end-to-end weakly supervised object detection network," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 8372–8381.

[28] P. Tang, X. Wang, X. Bai, and W. Liu, "Multiple instance detection network with online instance classifier refinement," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2843–2851.

[29] D. Li, J.-B. Huang, Y. Li, S. Wang, and M.-H. Yang, "Progressive representation adaptation for weakly supervised object localization," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 42, no. 6, pp. 1424–1438, Jun. 2020.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.

[30] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.

[31] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[32] X. Wei et al., "Fine-grained image analysis with deep learning: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 12, pp. 8927–8948, Dec. 2022.

[33] S. Branson, G. Van Horn, S. Belongie, and P. Perona, "Bird species categorization using pose normalized deep convolutional nets," 2014, *arXiv:1406.2952*.

[34] H. Zhang et al., "SPDA-CNN: Unifying semantic part detection and abstraction for fine-grained recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1143–1152.

[35] S. Huang, Z. Xu, D. Tao, and Y. Zhang, "Part-stacked CNN for fine-grained visual categorization," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 1173–1182.

[36] X.-S. Wei, C.-W. Xie, J. Wu, and C. Shen, "Mask-CNN: Localizing parts and selecting descriptors for fine-grained bird species categorization," *Pattern Recognit.*, vol. 76, pp. 704–714, Apr. 2018.

[37] D. Chen, Y. Zhong, A. Ma, Z. Zheng, and L. Zhang, "Explicable fine-grained aircraft recognition via deep part parsing prior framework for high-resolution remote sensing imagery," *IEEE Trans. Cybern.*, vol. 54, no. 7, pp. 3968–3979, Jul. 2024.

[38] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 5693–5703.

[39] M. Nauta, R. van Bree, and C. Seifert, "Neural prototype trees for interpretable fine-grained image recognition," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 14933–14943.

[40] J. Donnelly, A. J. Barnett, and C. Chen, "Deformable ProtoPNet: An interpretable image classifier using deformable prototypes," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 10265–10275.

[41] Y. Ukai, T. Hirakawa, T. Yamashita, and H. Fujiyoshi, "Toward prototypical part interpretable similarity learning with ProtoMetric," *IEEE Access*, vol. 11, pp. 62986–62997, 2023.

[42] J. Wang, P. Jiajie, Z. Liu, and H. Zhao, "HQProtoPnet: An evidence-based model for interpretable image recognition," in *Proc. Int. Joint Conf. Neural Netw. (IJCNN)*, Jun. 2023, pp. 1–8.

[43] B. Kim, R. Khanna, and O. O. Koyejo, "Examples are not enough, learn to criticize! criticism for interpretability," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–9.

[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[45] A. Gretton, K. M. Borgwardt, M. J. Rasch, B. Schölkopf, and A. Smola, "A kernel two-sample test," *J. Mach. Learn. Res.*, vol. 13, no. 1, pp. 723–773, Jan. 2012.

[46] T. A. Feo and M. G. Resende, "Greedy randomized adaptive search procedures," *J. Global Optim.*, vol. 6, no. 2, pp. 109–133, 1995.

[47] X. Liu, T. Xia, J. Wang, Y. Yang, F. Zhou, and Y. Lin, "Fully convolutional attention networks for fine-grained recognition," 2016, *arXiv:1603.06765*.

[48] K. He, X. Zhang, S. Ren, and J. Sun, "Spatial pyramid pooling in deep convolutional networks for visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 37, no. 9, pp. 1904–1916, Sep. 2015.

[49] O. Li, H. Liu, C. Chen, and C. Rudin, "Deep learning for case-based reasoning through prototypes: A neural network that explains its predictions," in *Proc. AAAI Conf. Artif. Intell.*, 2018, vol. 32, no. 1, pp. 1–8.

[50] Z. Zhang and M. Sabuncu, "Generalized cross entropy loss for training deep neural networks with noisy labels," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 31, 2018, pp. 1–11.

[51] L. Bottou, "Large-scale machine learning with stochastic gradient descent," in *Proc. 19th Int. Conf. Comput. Statist. (COMPSTAT)*, Paris, France. Heidelberg, Germany: Springer, 2010, pp. 177–186.

[52] J. Shermeyer, T. Hossler, A. V. Etten, D. Hogan, R. Lewis, and D. Kim, "RarePlanes: Synthetic data takes flight," in *Proc. IEEE Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2021, pp. 207–217.

[53] X. Sun et al., "FAIR1M: A benchmark dataset for fine-grained object recognition in high-resolution remote sensing imagery," *ISPRS J. Photogramm. Remote Sens.*, vol. 184, pp. 116–130, Feb. 2022.

[54] K. Simonyan and A. Zisserman, "Very deep convolutional networks for large-scale image recognition," 2014, *arXiv:1409.1556*.

[55] A. Howard et al., "Searching for MobileNetV3," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 1314–1324.

[56] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 116–131.

[57] K. Han, Y. Wang, Q. Tian, J. Guo, C. Xu, and C. Xu, "Ghostnet: More features from cheap operations," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2020, pp. 1580–1589.

[58] A. Dosovitskiy et al., "An image is worth 16×16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[59] Z. Liu et al., "Swin transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 10012–10022.

[60] T. Lin, A. RoyChowdhury, and S. Maji, "Bilinear convolutional neural networks for fine-grained visual recognition," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 6, pp. 1309–1322, Jun. 2018.

[61] G. Cheng, Q. Li, G. Wang, X. Xie, L. Min, and J. Han, "SFRNet: Fine-grained oriented object recognition via separate feature refinement," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610510.

[62] X. Huang, K. Xu, C. Huang, C. Wang, and K. Qin, "Multiple instance learning convolutional neural networks for fine-grained aircraft recognition," *Remote Sens.*, vol. 13, no. 24, p. 5132, Dec. 2021.



Zhengxi Guo received the B.S. degree from Hangzhou Dianzi University, Hangzhou, China, in 2020, where he is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China.

His research interests include deep learning, fine-grained aircraft classification, synthetic aperture radar (SAR) image interpretation and image understanding, and interpretability.



Biao Hou (Senior Member, IEEE) received the B.S. and M.S. degrees in mathematics from Northwest University, Xi'an, China, in 1996 and 1999, respectively, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2003.

Since 2003, he has been with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, School of Artificial Intelligence, Xidian University, where he is currently a Professor. His research interests include deep learning and synthetic aperture radar image interpretation.



Xianpeng Guo received the B.S. degree in intelligent science and technology and the Ph.D. degree in pattern recognition and intelligent systems from Xidian University, Xi'an, China, in 2015 and 2022, respectively.

He is currently a Lecturer with the School of Artificial Intelligence, Xidian University. His research interests include explainable artificial intelligence, deep learning, classification, and object detection in remote sensing images.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.



Zitong Wu received the B.S. and M.S. degrees from Northwest University, Xi'an, China, in 2015 and 2018, respectively, where he is currently pursuing the Ph.D. degree with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China.

His research interests include synthetic aperture radar image interpretation and deep learning in image processing.



Chen Yang received the B.S. and Ph.D. degrees in circuits and systems from Xidian University, Xi'an, China, in 2016 and 2022, respectively.

Since 2022, he has been a Lecturer with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education, Xidian University. His interests include the intelligent interpretation of polarimetric synthetic aperture radar (SAR) images, the design of physical mechanism deep networks, and terrain element extraction of remote sensing images.



Bo Ren (Member, IEEE) received the B.S. degree in telecommunications engineering from Northwest University, Xi'an, China, in 2011, and the Ph.D. degree in circuits and systems from Xidian University, Xi'an, in 2017.

In 2019, he joined Grenoble Images Parole Signal Automatique (GIPSA Lab), Grenoble, France, as a Visiting Scholar. Since 2022, he has been an Associate Professor with the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China, Xidian University. His research interests include data fusion, machine learning, and incremental learning in remote sensing images.



Licheng Jiao (Fellow, IEEE) was born in Shaanxi, China, in 1959. He received the B.S. degree from Shanghai Jiaotong University, Shanghai, China, in 1982, and the M.S. and Ph.D. degrees from Xi'an Jiaotong University, Xi'an, China, in 1984 and 1990, respectively.

From 1984 to 1986, he was an Assistant Professor with the Civil Aviation Institute of China, Tianjin, China. From 1990 to 1991, he was a Post-Doctoral Fellow with the Key Laboratory for Radar Signal Processing, Xidian University, Xi'an, where he is currently the Director of the Key Laboratory of Intelligent Perception and Image Understanding, Ministry of Education of China. His research interests include signal and image processing, nonlinear circuits and systems theory, learning theory and algorithms, optimization problems, wavelet theory, and machine learning.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 11:29:24 UTC from IEEE Xplore. Restrictions apply.